Functions

CS1010 Discrete Mathematics for Computer Science

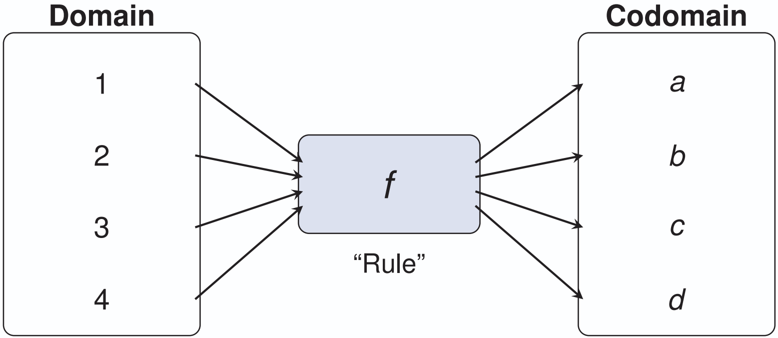
Shibam Ghosh

August 17, 2026

Department of CSE, IIT Hyderabad

Functions

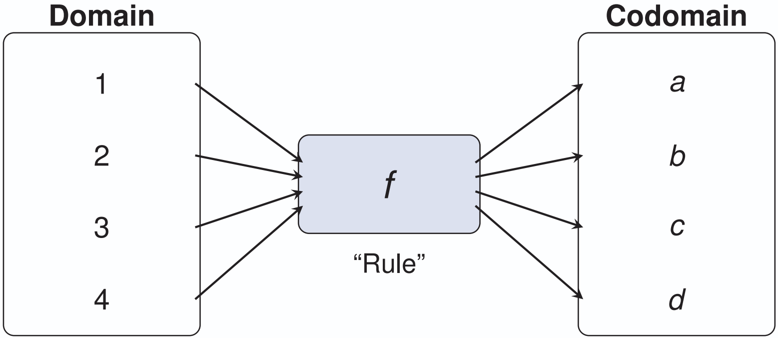
What is a Function?



Intuition

A function is a rule that takes an input and produces exactly one output.

What is a Function?



Intuition

A function is a rule that takes an input and produces exactly one output.

A Function is Like a Machine

Think of a Machine

- Put something in.
- The machine follows a rule.
- You get exactly one output.

$$f(x) = x^2$$

Input	Output
1	1
2	4
3	9
4	16

$$\boxed{\text{Input}} \longrightarrow \boxed{f} \longrightarrow \boxed{\text{Output}}$$

The Key Idea

Every input has one and only one output.

A Function is Like a Machine

Think of a Machine

- Put something in.
- The machine follows a rule.
- You get exactly one output.

$$f(x) = x^2$$

Input	Output
1	1
2	4
3	9
4	16

$$\boxed{\text{Input}} \longrightarrow \boxed{f} \longrightarrow \boxed{\text{Output}}$$

The Key Idea

Every input has one and only one output.

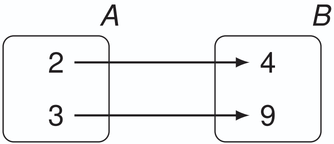
Formal Definition of a Function

Definition of function (mappings or transformations)

Let A and B be nonempty sets. A **function** f from A to B, denoted $f : A \to B$, is an assignment of exactly one element of B to each element of A.

The statement "exactly one" contains two requirements:

1. **Existence:** Every element of A must be assigned an output in B.
2. **Uniqueness:** No element of A can be assigned more than one output.



Notations

Given a function $f : A \to B$:

- **Domain:** Set A
- **Codomain:** Set B
- **Preimage:** If $f(a) = b$, a is the preimage of b.
- **Image:** If $f(a) = b$, b is the image of a.
- **Range:** The set of all images of elements in A. Denoted by $f(A)$.

$$f(A) = \{b \in B \mid \exists a \in A\,(f(a) = b)\} \subseteq B$$

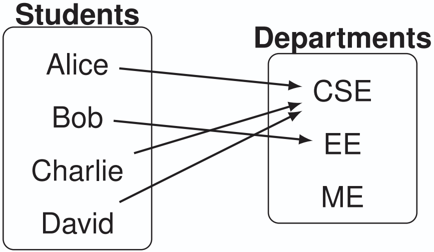
Function Equality

Two functions f and g are equal if they share the same domain, codomain, and mapping values for all domain elements.

Example

Let f : Students $\rightarrow$ Departments where each student is assigned to exactly one department.

- Domain: $\{\text{Alice}, \text{Bob}, \text{Charlie}, \text{David}\}$
- Codomain: $\{\text{CSE}, \text{EE}, \text{ME}\}$
- $$\begin{aligned} f(\text{Alice}) &= \text{CSE} \\ f(\text{Bob}) &= \text{EE} \\ f(\text{Charlie}) &= \text{CSE} \\ f(\text{David}) &= \text{CSE} \end{aligned}$$



- Range: $\{\text{CSE}, \text{EE}\}$
- Preimage of CSE: $\{\text{Alice}, \text{Charlie}, \text{David}\}$

Function Operations on Sets

Definition

Let $f : A \to B$ be a function, and $S \subseteq A$ be a subset of the domain. The **image of the subset** S under f, denoted by $f(S)$, is:

$$f(S) = \{b \in B \mid \exists a \in S\,(f(a) = b)\}$$

Exercise: From the previous example:

- Let $S = \{\text{Alice}, \text{Charlie}\}$. Find $f(S)$.
- Solution: $f(S) = \{f(\text{Alice}), f(\text{Charlie})\} = \{\text{CSE}\}$.

Injections (One-to-One) and Surjections (Onto)

Definition

A function $f : A \to B$ is **one-to-one** or **injective** if and only if distinct domain elements map to distinct codomain elements.

- $\forall a, \forall b,$ (if $f(a) = f(b)$ then $a = b$).
- Alternatively: $\forall a, \forall b,$ (if $a \neq b$ then $f(a) \neq f(b)$).

Definition

A function $f : A \to B$ is **onto** or **surjective** if and only if every element in the codomain B is mapped to by at least one element in the domain A.

- $\forall y \in B \exists x \in A (f(x) = y)$.
- Equivalently: $f(A) = B$.

Injections (One-to-One) and Surjections (Onto)

Definition

A function $f : A \to B$ is **one-to-one** or **injective** if and only if distinct domain elements map to distinct codomain elements.

- $\forall a, \forall b,$ (if $f(a) = f(b)$ then $a = b$).
- Alternatively: $\forall a, \forall b,$ (if $a \neq b$ then $f(a) \neq f(b)$).

Definition

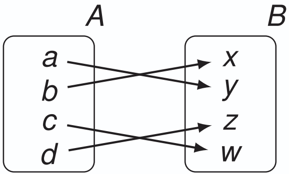
A function $f : A \to B$ is **onto** or **surjective** if and only if every element in the codomain B is mapped to by at least one element in the domain A.

- $\forall y \in B \exists x \in A (f(x) = y)$.
- Equivalently: $f(A) = B$.

Bijections (One-to-One Correspondence)

Definition

A function f is a **bijection** (one-to-one correspondence) if it is both injective and surjective (both one-to-one and onto).



Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{N} \to \mathbb{N}$, $f(n) = 2n$. Is f injective? Is it surjective?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{N} \to \mathbb{N}, f(n) = 2n$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **injective** but not **surjective**.

- Injective: Since $f(a) = f(b) \Rightarrow 2a = 2b \Rightarrow a = b$.
- Surjective: No, because odd numbers (e.g., $1, 3, 5, \ldots$) are never images.

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{N} \to \mathbb{N}, f(n) = 2n$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **injective** but not **surjective**.

- Injective: Since $f(a) = f(b) \Rightarrow 2a = 2b \Rightarrow a = b$.
- Surjective: No, because odd numbers (e.g., $1, 3, 5, \ldots$) are never images.

$\mathcal{Q}$. Let $f : \mathbb{Z} \to \mathbb{N}_0, f(x) = |x|$. Is f injective? Is it surjective?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{N} \to \mathbb{N}, f(n) = 2n$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **injective** but not **surjective**.

- Injective: Since $f(a) = f(b) \Rightarrow 2a = 2b \Rightarrow a = b$.
- Surjective: No, because odd numbers (e.g., $1, 3, 5, \ldots$) are never images.

$\mathcal{Q}$. Let $f : \mathbb{Z} \to \mathbb{N}_0, f(x) = |x|$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **surjective** but not **injective**.

- Injective: No, since $f(2) = f(-2) = 2$.
- Surjective: Yes, every non-negative integer has a preimage.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x + 5$. Is f injective? Is it surjective?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{N} \to \mathbb{N}, f(n) = 2n$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **injective** but not **surjective**.

- Injective: Since $f(a) = f(b) \Rightarrow 2a = 2b \Rightarrow a = b$.
- Surjective: No, because odd numbers (e.g., $1, 3, 5, \ldots$) are never images.

$\mathcal{Q}$. Let $f : \mathbb{Z} \to \mathbb{N}_0, f(x) = |x|$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **surjective** but not **injective**.

- Injective: No, since $f(2) = f(-2) = 2$.
- Surjective: Yes, every non-negative integer has a preimage.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x + 5$. Is f injective? Is it surjective?

$\mathcal{A}$. f is both **injective and surjective**, hence bijective.

- For injectivity, $f(a) = f(b) \Rightarrow a + 5 = b + 5 \Rightarrow a = b$.
- For surjectivity, given any $y \in \mathbb{R}$, choose $x = y - 5$. Then $f(x) = y$.

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 + 1$. Is f injective? Is it surjective?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2 + 1$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **neither injective nor surjective.**

- Not injective as $f(1) = f(-1) = 2$,.
- Not surjective as $f(x) \geq 1$ for every $x \in \mathbb{R}$. Thus, e.g., 0 has no preimage.

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2 + 1$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **neither injective nor surjective.**

- Not injective as $f(1) = f(-1) = 2,$.
- Not surjective as $f(x) \geq 1$ for every $x \in \mathbb{R}$. Thus, e.g., 0 has no preimage.

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$. Is f surjective?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2 + 1$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **neither injective nor surjective.**

- Not injective as $f(1) = f(-1) = 2$,.
- Not surjective as $f(x) \geq 1$ for every $x \in \mathbb{R}$. Thus, e.g., 0 has no preimage.

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$. Is f surjective?

$\mathcal{A}$. No. Negative real numbers are not in the image of f.

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2 + 1$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **neither injective nor surjective.**

- Not injective as $f(1) = f(-1) = 2$,.
- Not surjective as $f(x) \geq 1$ for every $x \in \mathbb{R}$. Thus, e.g., 0 has no preimage.

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$. Is f surjective?

$\mathcal{A}$. No. Negative real numbers are not in the image of f.

$\mathcal{Q}$. Now consider $g : \mathbb{R} \to [0, \infty), g(x) = x^2$. Is g surjective?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2 + 1$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **neither injective nor surjective.**

- Not injective as $f(1) = f(-1) = 2$,.
- Not surjective as $f(x) \geq 1$ for every $x \in \mathbb{R}$. Thus, e.g., 0 has no preimage.

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$. Is f surjective?

$\mathcal{A}$. No. Negative real numbers are not in the image of f.

$\mathcal{Q}$. Now consider $g : \mathbb{R} \to [0, \infty), g(x) = x^2$. Is g surjective?

$\mathcal{A}$. Yes. For every $y \in [0, \infty)$, choose $x = \sqrt{y}$. Then $g(x) = y$.

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2 + 1$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **neither injective nor surjective.**

- Not injective as $f(1) = f(-1) = 2$,.
- Not surjective as $f(x) \geq 1$ for every $x \in \mathbb{R}$. Thus, e.g., 0 has no preimage.

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$. Is f surjective?

$\mathcal{A}$. No. Negative real numbers are not in the image of f.

$\mathcal{Q}$. Now consider $g : \mathbb{R} \to [0, \infty), g(x) = x^2$. Is g surjective?

$\mathcal{A}$. Yes. For every $y \in [0, \infty)$, choose $x = \sqrt{y}$. Then $g(x) = y$.

$\mathcal{Q}$. The formulas are identical. Why is one function surjective and the other not?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2 + 1$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **neither injective nor surjective.**

- Not injective as $f(1) = f(-1) = 2$,.
- Not surjective as $f(x) \geq 1$ for every $x \in \mathbb{R}$. Thus, e.g., 0 has no preimage.

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$. Is f surjective?

$\mathcal{A}$. No. Negative real numbers are not in the image of f.

$\mathcal{Q}$. Now consider $g : \mathbb{R} \to [0, \infty), g(x) = x^2$. Is g surjective?

$\mathcal{A}$. Yes. For every $y \in [0, \infty)$, choose $x = \sqrt{y}$. Then $g(x) = y$.

$\mathcal{Q}$. The formulas are identical. Why is one function surjective and the other not?

$\mathcal{A}$. Because the **codomains are different.**

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $A = \{1, 2, 3\}, B = \{a, b, c, d\}$. Can there be a surjective function $f : A \to B$?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $A = \{1, 2, 3\}, B = \{a, b, c, d\}$. Can there be a surjective function $f : A \to B$?

$\mathcal{A}$. No. There are only 3 elements in the domain but 4 elements in the codomain. At least one element of B must remain unused.

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $A = \{1, 2, 3\}, B = \{a, b, c, d\}$. Can there be a surjective function $f : A \to B$?

$\mathcal{A}$. No. There are only 3 elements in the domain but 4 elements in the codomain. At least one element of B must remain unused.

$\mathcal{Q}$. Can there be an injective function $f : A \to B$?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $A = \{1, 2, 3\}, B = \{a, b, c, d\}$. Can there be a surjective function $f : A \to B$?

$\mathcal{A}$. No. There are only 3 elements in the domain but 4 elements in the codomain. At least one element of B must remain unused.

$\mathcal{Q}$. Can there be an injective function $f : A \to B$?

$\mathcal{A}$. Yes. For example, $1 \mapsto a, 2 \mapsto b, 3 \mapsto c$.

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $A = \{1, 2, 3\}, B = \{a, b, c, d\}$. Can there be a surjective function $f : A \to B$?

$\mathcal{A}$. No. There are only 3 elements in the domain but 4 elements in the codomain. At least one element of B must remain unused.

$\mathcal{Q}$. Can there be an injective function $f : A \to B$?

$\mathcal{A}$. Yes. For example, $1 \mapsto a, 2 \mapsto b, 3 \mapsto c$.

$\mathcal{Q}$. Can there be a bijection $f : A \to B$?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $A = \{1, 2, 3\}, B = \{a, b, c, d\}$. Can there be a surjective function $f : A \to B$?

$\mathcal{A}$. No. There are only 3 elements in the domain but 4 elements in the codomain. At least one element of B must remain unused.

$\mathcal{Q}$. Can there be an injective function $f : A \to B$?

$\mathcal{A}$. Yes. For example, $1 \mapsto a, 2 \mapsto b, 3 \mapsto c$.

$\mathcal{Q}$. Can there be a bijection $f : A \to B$?

$\mathcal{A}$. No. A bijection requires the domain and codomain to have the same number of elements.

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $A = \{1, 2, 3\}, B = \{a, b, c, d\}$. Can there be a surjective function $f : A \to B$?

$\mathcal{A}$. No. There are only 3 elements in the domain but 4 elements in the codomain. At least one element of B must remain unused.

$\mathcal{Q}$. Can there be an injective function $f : A \to B$?

$\mathcal{A}$. Yes. For example, $1 \mapsto a, 2 \mapsto b, 3 \mapsto c$.

$\mathcal{Q}$. Can there be a bijection $f : A \to B$?

$\mathcal{A}$. No. A bijection requires the domain and codomain to have the same number of elements.

$\mathcal{Q}$. Let $f : \mathbb{Z} \to \mathbb{Z}, f(x) = 2x + 1$. Is f injective? Is it surjective?

Quick Check: Injective or Surjective?

$\mathcal{Q}$. Let $A = \{1, 2, 3\}, B = \{a, b, c, d\}$. Can there be a surjective function $f : A \to B$?

$\mathcal{A}$. No. There are only 3 elements in the domain but 4 elements in the codomain. At least one element of B must remain unused.

$\mathcal{Q}$. Can there be an injective function $f : A \to B$?

$\mathcal{A}$. Yes. For example, $1 \mapsto a, 2 \mapsto b, 3 \mapsto c$.

$\mathcal{Q}$. Can there be a bijection $f : A \to B$?

$\mathcal{A}$. No. A bijection requires the domain and codomain to have the same number of elements.

$\mathcal{Q}$. Let $f : \mathbb{Z} \to \mathbb{Z}, f(x) = 2x + 1$. Is f injective? Is it surjective?

$\mathcal{A}$. f is **injective but not surjective.**

- If $2a + 1 = 2b + 1$, then $a = b$, so f is injective.
- Every output of f is odd. So even integers has no preimage.

Quick Check: Is It a Function?

Determine whether f is a function from the set of all bit strings to the set of integers.

$\mathfrak{Q}$. $f(S)$ is the position of a 0 bit in S.

Quick Check: Is It a Function?

Determine whether f is a function from the set of all bit strings to the set of integers.

$\mathcal{Q}$. $f(S)$ is the position of a 0 bit in S.

$\mathcal{A}$. **No.** The position of a 0 bit need not be unique. For example, for $S = 10100$, the positions of the 0 bits are 2, 4, and 5.

Quick Check: Is It a Function?

Determine whether f is a function from the set of all bit strings to the set of integers.

$\mathcal{Q}$. $f(S)$ is the position of a 0 bit in S.

$\mathcal{A}$. **No.** The position of a 0 bit need not be unique. For example, for $S = 10100$, the positions of the 0 bits are 2, 4, and 5.

$\mathcal{Q}$. $f(S)$ is the number of 1 bits in S.

Quick Check: Is It a Function?

Determine whether f is a function from the set of all bit strings to the set of integers.

$\mathcal{Q}$. $f(S)$ is the position of a 0 bit in S.

$\mathcal{A}$. **No.** The position of a 0 bit need not be unique. For example, for $S = 10100$, the positions of the 0 bits are 2, 4, and 5.

$\mathcal{Q}$. $f(S)$ is the number of 1 bits in S.

$\mathcal{A}$. **Yes.** Every bit string has a unique number of 1 bits. For example, $f(101101) = 4$.

Quick Check: Is It a Function?

Determine whether f is a function from the set of all bit strings to the set of integers.

$\mathcal{Q}$. $f(S)$ is the position of a 0 bit in S.

$\mathcal{A}$. **No.** The position of a 0 bit need not be unique. For example, for $S = 10100$, the positions of the 0 bits are 2, 4, and 5.

$\mathcal{Q}$. $f(S)$ is the number of 1 bits in S.

$\mathcal{A}$. **Yes.** Every bit string has a unique number of 1 bits. For example, $f(101101) = 4$.

$\mathcal{Q}$. $f(S)$ is the smallest integer i such that the ith bit of S is 1, and $f(S) = 0$ when S is the empty string.

Quick Check: Is It a Function?

Determine whether f is a function from the set of all bit strings to the set of integers.

$\mathfrak{Q}$. $f(S)$ is the position of a 0 bit in S.

$\mathcal{A}$. **No.** The position of a 0 bit need not be unique. For example, for $S = 10100$, the positions of the 0 bits are 2, 4, and 5.

$\mathfrak{Q}$. $f(S)$ is the number of 1 bits in S.

$\mathcal{A}$. **Yes.** Every bit string has a unique number of 1 bits. For example, $f(101101) = 4$.

$\mathfrak{Q}$. $f(S)$ is the smallest integer i such that the ith bit of S is 1, and $f(S) = 0$ when S is the empty string.

$\mathcal{A}$. **No.** Consider a nonempty string consisting entirely of 0's, such as $S = 000$. There is no position i for which the ith bit is 1, and no value for $f(S)$ is specified.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = m^2 + n^2$.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = m^2 + n^2$.

$\mathcal{A}$. **No.** Since $m^2 + n^2 \geq 0$, no negative integer can be an output. For example, -1 has no preimage.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = m^2 + n^2$.

$\mathcal{A}$. **No.** Since $m^2 + n^2 \geq 0$, no negative integer can be an output. For example, -1 has no preimage.

$\mathcal{Q}$. $f(m, n) = m$.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = m^2 + n^2$.

$\mathcal{A}$. **No.** Since $m^2 + n^2 \geq 0$, no negative integer can be an output. For example, -1 has no preimage.

$\mathcal{Q}$. $f(m, n) = m$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and any $n \in \mathbb{Z}$, for example $n = 0$. Then $f(m, n) = y$.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = m^2 + n^2$.

$\mathcal{A}$. **No.** Since $m^2 + n^2 \geq 0$, no negative integer can be an output. For example, -1 has no preimage.

$\mathcal{Q}$. $f(m, n) = m$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and any $n \in \mathbb{Z}$, for example $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = |n|$.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = m^2 + n^2$.

$\mathcal{A}$. **No.** Since $m^2 + n^2 \geq 0$, no negative integer can be an output. For example, -1 has no preimage.

$\mathcal{Q}$. $f(m, n) = m$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and any $n \in \mathbb{Z}$, for example $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = |n|$.

$\mathcal{A}$. **No.** The output is always non-negative, so no negative integer is in the range.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = m^2 + n^2$.

$\mathcal{A}$. **No.** Since $m^2 + n^2 \geq 0$, no negative integer can be an output. For example, -1 has no preimage.

$\mathcal{Q}$. $f(m, n) = m$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and any $n \in \mathbb{Z}$, for example $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = |n|$.

$\mathcal{A}$. **No.** The output is always non-negative, so no negative integer is in the range.

$\mathcal{Q}$. **(e)** $f(m, n) = m - n$.

Quick Check: Onto or Not?

Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto.

$\mathcal{Q}$. $f(m, n) = m + n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = m^2 + n^2$.

$\mathcal{A}$. **No.** Since $m^2 + n^2 \geq 0$, no negative integer can be an output. For example, -1 has no preimage.

$\mathcal{Q}$. $f(m, n) = m$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and any $n \in \mathbb{Z}$, for example $n = 0$. Then $f(m, n) = y$.

$\mathcal{Q}$. $f(m, n) = |n|$.

$\mathcal{A}$. **No.** The output is always non-negative, so no negative integer is in the range.

$\mathcal{Q}$. **(e)** $f(m, n) = m - n$.

$\mathcal{A}$. **Yes.** Given any $y \in \mathbb{Z}$, choose $m = y$ and $n = 0$. Then $f(m, n) = y$.

Quick Check: Is the Function a Bijection?

Determine whether each function is a bijection from $\mathbb{R}$ to $\mathbb{R}$.

$\mathcal{Q}$. $f(x) = x^3$.

Quick Check: Is the Function a Bijection?

Determine whether each function is a bijection from $\mathbb{R}$ to $\mathbb{R}$.

$\mathcal{Q}$. $f(x) = x^3$.

$\mathcal{A}$. **Yes.**

- It is injective since $a^3 = b^3 \Rightarrow a = b$.
- It is surjective since, for any $y \in \mathbb{R}$, choosing $x = \sqrt[3]{y}$ gives $f(x) = y$.

Quick Check: Is the Function a Bijection?

Determine whether each function is a bijection from $\mathbb{R}$ to $\mathbb{R}$.

$\mathcal{Q}$. $f(x) = x^3$.

$\mathcal{A}$. **Yes.**

- It is injective since $a^3 = b^3 \Rightarrow a = b$.
- It is surjective since, for any $y \in \mathbb{R}$, choosing $x = \sqrt[3]{y}$ gives $f(x) = y$.

$\mathcal{Q}$. $f(x) = \frac{x^2+1}{x^2+2}$.

Quick Check: Is the Function a Bijection?

Determine whether each function is a bijection from $\mathbb{R}$ to $\mathbb{R}$.

$\mathcal{Q}$. $f(x) = x^3$.

$\mathcal{A}$. **Yes.**

- It is injective since $a^3 = b^3 \Rightarrow a = b$.
- It is surjective since, for any $y \in \mathbb{R}$, choosing $x = \sqrt[3]{y}$ gives $f(x) = y$.

$\mathcal{Q}$. $f(x) = \frac{x^2+1}{x^2+2}$.

$\mathcal{A}$. **No.**

- It is not injective since $f(x) = f(-x)$ for every x, with $x \neq -x$ when $x \neq 0$.
- It is also not surjective: since $x^2 \geq 0$, $\frac{1}{2} \leq f(x) < 1$. Thus its range is contained in $[1/2, 1)$, not all of $\mathbb{R}$.

Some Important Functions

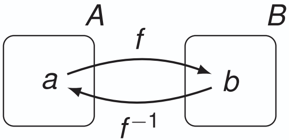
Inverse Functions

Definition

Let $f : A \to B$ be a bijection. The **inverse function** of f, denoted $f^{-1} : B \to A$, maps $b \in B$ to the unique element $a \in A$ such that $f(a) = b$.

$$f^{-1}(b) = a \iff f(a) = b$$

No inverse function exists unless f is a bijection



Why Must f Be a Bijection to Have an Inverse?

The requirement that f be bijective is essential.

- If f is **not injective**, two different elements of A may have the same image. Then the inverse would not know which element of A to choose.
- If f is **not surjective**, some element of B has no preimage. Then the inverse would not be defined on that element.

Therefore, an inverse function

$$f^{-1} : B \to A$$

exists precisely when $f : A \to B$ is bijective.

The Indicator Function

Definition

The **indicator function** (or characteristic function) of a subset A of the set X is a function $\mathbf{1}_A : X \to \{0, 1\}$ defined as:

$$\mathbf{1}_A(x) := \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Example: Vowel Detection in Text Processing

Let $X = \{c, o, d, e\}$ and A be the subset of all English vowels: $A = \{a, e, i, o, u\}$. The indicator function $\mathbf{1}_A(x)$ scans our set X and maps each character:

- $\mathbf{1}_A(c) = 0 \quad (\text{since } c \notin A)$
- $\mathbf{1}_A(o) = 1 \quad (\text{since } o \in A)$
- $\mathbf{1}_A(d) = 0 \quad (\text{since } d \notin A)$
- $\mathbf{1}_A(e) = 1 \quad (\text{since } e \in A)$

Indicator Functions and Set Operations

Let $A, B \subseteq X$ and let $\mathbf{1}_A, \mathbf{1}_B : X \to \{0, 1\}$ be their indicator functions.

- **Intersection:**

$$\boxed{\mathbf{1}_{A\cap B}(x) = \mathbf{1}_A(x)\mathbf{1}_B(x)}$$

The product is 1 exactly when both $x \in A$ and $x \in B$.

- **Complement:**

$$\boxed{\mathbf{1}_{\overline{A}}(x) = 1 - \mathbf{1}_A(x)}$$

If $x \in A$, the value changes from 1 to 0; otherwise it changes from 0 to 1.

- **Union:**

$$\boxed{\mathbf{1}_{A\cup B}(x) = \mathbf{1}_A(x) + \mathbf{1}_B(x) - \mathbf{1}_{A\cap B}(x)}$$

We subtract the intersection because its elements were counted twice.

Indicator Functions and Set Operations

Let $A, B \subseteq X$ and let $\mathbf{1}_A, \mathbf{1}_B : X \to \{0, 1\}$ be their indicator functions.

- **Intersection:**

$$\boxed{\mathbf{1}_{A\cap B}(x) = \mathbf{1}_A(x)\mathbf{1}_B(x)}$$

  The product is 1 exactly when both $x \in A$ and $x \in B$.

- **Complement:**

$$\boxed{\mathbf{1}_{\overline{A}}(x) = 1 - \mathbf{1}_A(x)}$$

  If $x \in A$, the value changes from 1 to 0; otherwise it changes from 0 to 1.

- **Union:**

$$\boxed{\mathbf{1}_{A\cup B}(x) = \mathbf{1}_A(x) + \mathbf{1}_B(x) - \mathbf{1}_{A\cap B}(x)}$$

  We subtract the intersection because its elements were counted twice.

Indicator Functions and Set Operations

Let $A, B \subseteq X$ and let $\mathbf{1}_A, \mathbf{1}_B : X \to \{0, 1\}$ be their indicator functions.

- **Intersection:**

$$\boxed{\mathbf{1}_{A\cap B}(x) = \mathbf{1}_A(x)\mathbf{1}_B(x)}$$

  The product is 1 exactly when both $x \in A$ and $x \in B$.

- **Complement:**

$$\boxed{\mathbf{1}_{\overline{A}}(x) = 1 - \mathbf{1}_A(x)}$$

  If $x \in A$, the value changes from 1 to 0; otherwise it changes from 0 to 1.

- **Union:**

$$\boxed{\mathbf{1}_{A\cup B}(x) = \mathbf{1}_A(x) + \mathbf{1}_B(x) - \mathbf{1}_{A\cap B}(x)}$$

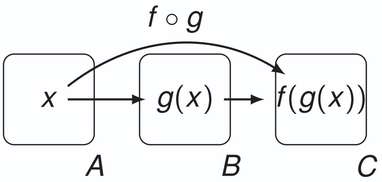
  We subtract the intersection because its elements were counted twice.

Composition of Functions

Definition

Let $g : A \to B$ and $f : B \to C$. The **composition** of f with g, denoted by $f \circ g$, is the function from A to C defined by:

$$(f \circ g)(x) = f(g(x))$$



Remember the Order: In $(f \circ g)(x) = f(g(x))$, g **is applied first**, followed by f.

Composition: When Is It Defined?

The Key Requirement

To form $f \circ g$, the output of g must be a valid input for f.

Suppose $g : A \to B and f : B \to C$.

Then the output of g lies in B, which is exactly the domain of f. Therefore, the composition is well-defined: $\boxed{f \circ g : A \to C}$.

$$\begin{array}{ccccc} A & & B & & C \\ \boxed{x} & \xrightarrow{g} & \boxed{g(x)} & \xrightarrow{f} & \boxed{f(g(x))} \end{array}$$

Quick Check: Inverse Functions

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x + 2$. Is f invertible? If so, find f^{-1}.

Quick Check: Inverse Functions

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = 3x + 2$. Is f invertible? If so, find f^{-1}.

Solution:

$$y = 3x + 2 \quad \Longrightarrow \quad x = \frac{y-2}{3}. \text{ Hence, } \boxed{f^{-1}(y) = \frac{y-2}{3}}.$$

Quick Check: Inverse Functions

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = 3x + 2$. Is f invertible? If so, find f^{-1}.

Solution:

$$y = 3x + 2 \quad \Longrightarrow \quad x = \frac{y-2}{3}. \text{ Hence, } \boxed{f^{-1}(y) = \frac{y-2}{3}}.$$

$\mathcal{Q}$. Verify that your answer is correct by computing $f^{-1}(f(x))$.

Quick Check: Inverse Functions

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x + 2$. Is f invertible? If so, find f^{-1}.

Solution:

$$y = 3x + 2 \implies x = \frac{y-2}{3}. \text{ Hence, } \boxed{f^{-1}(y) = \frac{y-2}{3}}.$$

$\mathcal{Q}$. Verify that your answer is correct by computing $f^{-1}(f(x))$.

$\mathcal{A}$. $f^{-1}(f(x)) = f^{-1}(3x+2) = \frac{(3x+2)-2}{3} = x$.

Quick Check: Inverse Functions

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x + 2$. Is f invertible? If so, find f^{-1}.

Solution:

$$y = 3x + 2 \quad \Longrightarrow \quad x = \frac{y-2}{3}. \text{ Hence, } \boxed{f^{-1}(y) = \frac{y-2}{3}}.$$

$\mathcal{Q}$. Verify that your answer is correct by computing $f^{-1}(f(x))$.

$\mathcal{A}$. $f^{-1}(f(x)) = f^{-1}(3x+2) = \frac{(3x+2)-2}{3} = x.$

$\mathcal{Q}$. Can we modify the domain so that x^2 becomes invertible?

Quick Check: Inverse Functions

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x + 2$. Is f invertible? If so, find f^{-1}.

Solution:

$$y = 3x + 2 \quad \Longrightarrow \quad x = \frac{y-2}{3}. \text{ Hence, } \boxed{f^{-1}(y) = \frac{y-2}{3}}.$$

$\mathcal{Q}$. Verify that your answer is correct by computing $f^{-1}(f(x))$.

$\mathcal{A}$. $f^{-1}(f(x)) = f^{-1}(3x+2) = \frac{(3x+2)-2}{3} = x$.

$\mathcal{Q}$. Can we modify the domain so that x^2 becomes invertible?

$\mathcal{A}$. Yes. Consider $f : [0, \infty) \to [0, \infty)$, $\quad f(x) = x^2$. This function is bijective, and therefore $f^{-1}(x) = \sqrt{x}$.

Quick Check: Invertibility and Codomain

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}$, $f(x) = e^x$. Show that f is not invertible. What happens if we change the codomain to $\mathbb{R}^+$?

Quick Check: Invertibility and Codomain

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = e^x$. Show that f is not invertible. What happens if we change the codomain to $\mathbb{R}^+$?

$\mathcal{A}$. **As a function $\mathbb{R} \to \mathbb{R}$:** f is not surjective. Since $e^x > 0$ for every $x \in \mathbb{R}$, no negative real number or 0 occurs as an output. Hence, $f(\mathbb{R}) = (0, \infty) \neq \mathbb{R}$. Therefore, f is not bijective and has no inverse $f^{-1} : \mathbb{R} \to \mathbb{R}$.

Quick Check: Invertibility and Codomain

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = e^x$. Show that f is not invertible. What happens if we change the codomain to $\mathbb{R}^+$?

$\mathcal{A}$. **As a function $\mathbb{R} \to \mathbb{R}$:** f is not surjective. Since $e^x > 0$ for every $x \in \mathbb{R}$, no negative real number or 0 occurs as an output. Hence, $f(\mathbb{R}) = (0, \infty) \neq \mathbb{R}$. Therefore, f is not bijective and has no inverse $f^{-1} : \mathbb{R} \to \mathbb{R}$.

$\mathcal{Q}$. Now consider $g : \mathbb{R} \to \mathbb{R}^+, \quad g(x) = e^x$. Is g invertible?

Quick Check: Invertibility and Codomain

$\mathcal{Q}$. Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = e^x$. Show that f is not invertible. What happens if we change the codomain to $\mathbb{R}^+$?

$\mathcal{A}$. **As a function $\mathbb{R} \to \mathbb{R}$:** f is not surjective. Since $e^x > 0$ for every $x \in \mathbb{R}$, no negative real number or 0 occurs as an output. Hence, $f(\mathbb{R}) = (0, \infty) \neq \mathbb{R}$. Therefore, f is not bijective and has no inverse $f^{-1} : \mathbb{R} \to \mathbb{R}$.

$\mathcal{Q}$. Now consider $g : \mathbb{R} \to \mathbb{R}^+, \quad g(x) = e^x$. Is g invertible?

$\mathcal{A}$. **Yes.** The function is injective because $e^a = e^b \implies a = b$. It is also surjective onto $\mathbb{R}^+$ because for every $y > 0$, choosing $x = \ln y$ gives $g(x) = e^{\ln y} = y$.

Quick Check: Function Composition

In each question, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x + 2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = 3x$.

Quick Check: Function Composition

In each question, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x + 2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = 3x$.

$\mathcal{A}$. $(f \circ g)(x) = f(3x) = 3x + 2$, and $(g \circ f)(x) = g(x + 2) = 3(x + 2) = 3x + 6$.

Quick Check: Function Composition

In each question, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x + 2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = 3x$.

$\mathcal{A}$. $(f \circ g)(x) = f(3x) = 3x + 2$, and $(g \circ f)(x) = g(x + 2) = 3(x + 2) = 3x + 6$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = x + 1$.

Quick Check: Function Composition

In each question, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x + 2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = 3x$.

$\mathcal{A}$. $(f \circ g)(x) = f(3x) = 3x + 2$, and $(g \circ f)(x) = g(x + 2) = 3(x + 2) = 3x + 6$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = x + 1$.

$\mathcal{A}$. $(f \circ g)(x) = f(x + 1) = (x + 1)^2$, and $(g \circ f)(x) = g(x^2) = x^2 + 1$.

Quick Check: Function Composition

In each question, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x + 2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = 3x$.

$\mathcal{A}$. $(f \circ g)(x) = f(3x) = 3x + 2$, and $(g \circ f)(x) = g(x + 2) = 3(x + 2) = 3x + 6$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = x + 1$.

$\mathcal{A}$. $(f \circ g)(x) = f(x + 1) = (x + 1)^2$, and $(g \circ f)(x) = g(x^2) = x^2 + 1$.

Composition is generally **not commutative**; the order matters. $\boxed{f \circ g \neq g \circ f}$

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2,$ and $g : \{a, b, c\} \to \mathbb{R}$. Are $f \circ g$ and $g \circ f$ defined?

Quick Check: Function Composition

In each question, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x + 2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = 3x$.

$\mathcal{A}$. $(f \circ g)(x) = f(3x) = 3x + 2$, and $(g \circ f)(x) = g(x + 2) = 3(x + 2) = 3x + 6$.

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = x + 1$.

$\mathcal{A}$. $(f \circ g)(x) = f(x + 1) = (x + 1)^2$, and $(g \circ f)(x) = g(x^2) = x^2 + 1$.

Composition is generally **not commutative**; the order matters. $\boxed{f \circ g \neq g \circ f}$

$\mathcal{Q}$. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2,$ and $g : \{a, b, c\} \to \mathbb{R}$. Are $f \circ g$ and $g \circ f$ defined?

$\mathcal{A}$. $f \circ g$ **is defined** because g maps into $\mathbb{R}$, which is the domain of f.

$\mathcal{A}$. $g \circ f$ **is not defined** because f maps into $\mathbb{R}$, while the domain of g is $\{a, b, c\}$.

Composition and Bijections

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. Suppose that f and g are both one-to-one. Show that $f \circ g$ is one-to-one.

$\mathcal{A}$. Let $x, y \in A$ with $x \neq y$.

- Since g is one-to-one, $g(x) \neq g(y)$.
- Since f is one-to-one, $f(g(x)) \neq f(g(y))$.
- Therefore, $f \circ g$ is one-to-one as $\boxed{(f \circ g)(x) \neq (f \circ g)(y)}$,

$\mathcal{Q}$. Suppose that f and g are both onto. Show that $f \circ g$ is onto.

$\mathcal{A}$. Let $y \in C$ be arbitrary.

- Since f is onto, there exists $b \in B$ such that $f(b) = y$.
- Since g is onto, there exists $x \in A$ such that $g(x) = b$.
- Therefore, $(f \circ g)(x) = f(g(x)) = f(b) = y$. Hence every $y \in C$ has a preimage in A.

Composition and Bijections

Let $g : A \to B$ and $f : B \to C$.

$\mathfrak{Q}$. Suppose that f and g are both one-to-one. Show that $f \circ g$ is one-to-one.

$\mathcal{A}$. Let $x, y \in A$ with $x \neq y$.

- Since g is one-to-one, $g(x) \neq g(y)$.
- Since f is one-to-one, $f(g(x)) \neq f(g(y))$.
- Therefore, $f \circ g$ is one-to-one as $\boxed{(f \circ g)(x) \neq (f \circ g)(y)}$,

$\mathfrak{Q}$. Suppose that f and g are both onto. Show that $f \circ g$ is onto.

$\mathcal{A}$. Let $y \in C$ be arbitrary.

- Since f is onto, there exists $b \in B$ such that $f(b) = y$.
- Since g is onto, there exists $x \in A$ such that $g(x) = b$.
- Therefore, $(f \circ g)(x) = f(g(x)) = f(b) = y$. Hence every $y \in C$ has a preimage in A.

Composition and Bijections

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. Suppose that f and g are both one-to-one. Show that $f \circ g$ is one-to-one.

$\mathcal{A}$. Let $x, y \in A$ with $x \neq y$.

- Since g is one-to-one, $g(x) \neq g(y)$.
- Since f is one-to-one, $f(g(x)) \neq f(g(y))$.
- Therefore, $f \circ g$ is one-to-one as $\boxed{(f \circ g)(x) \neq (f \circ g)(y)}$,

$\mathcal{Q}$. Suppose that f and g are both onto. Show that $f \circ g$ is onto.

$\mathcal{A}$. Let $y \in C$ be arbitrary.

- Since f is onto, there exists $b \in B$ such that $f(b) = y$.
- Since g is onto, there exists $x \in A$ such that $g(x) = b$.
- Therefore, $(f \circ g)(x) = f(g(x)) = f(b) = y$. Hence every $y \in C$ has a preimage in A.

Composition and Bijections

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. Suppose that f and g are both one-to-one. Show that $f \circ g$ is one-to-one.

$\mathcal{A}$. Let $x, y \in A$ with $x \neq y$.

- Since g is one-to-one, $g(x) \neq g(y)$.
- Since f is one-to-one, $f(g(x)) \neq f(g(y))$.
- Therefore, $f \circ g$ is one-to-one as $\boxed{(f \circ g)(x) \neq (f \circ g)(y)}$,

$\mathcal{Q}$. Suppose that f and g are both onto. Show that $f \circ g$ is onto.

$\mathcal{A}$. Let $y \in C$ be arbitrary.

- Since f is onto, there exists $b \in B$ such that $f(b) = y$.
- Since g is onto, there exists $x \in A$ such that $g(x) = b$.
- Therefore, $(f \circ g)(x) = f(g(x)) = f(b) = y$. Hence every $y \in C$ has a preimage in A.

Composition and Surjectivity

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. Suppose that $f \circ g$ is onto. Prove that f must also be onto.

$\mathcal{A}$. Let $y \in C$ be arbitrary. Since $f \circ g$ is onto, there exists some $x \in A$ such that

$$(f \circ g)(x) = y. \implies f(g(x)) = y.$$

Since $g(x) \in B$, we have found an element of B, namely $g(x)$, that is mapped by f to y. Thus, every $y \in C$ has a preimage under f. Hence,

$$\boxed{f \text{ is onto.}}$$

Composition and Surjectivity

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. Suppose that $f \circ g$ is onto. Prove that f must also be onto.

$\mathcal{A}$. Let $y \in C$ be arbitrary. Since $f \circ g$ is onto, there exists some $x \in A$ such that

$$(f \circ g)(x) = y. \implies f(g(x)) = y.$$

Since $g(x) \in B$, we have found an element of B, namely $g(x)$, that is mapped by f to y. Thus, every $y \in C$ has a preimage under f. Hence,

$$\boxed{f \text{ is onto.}}$$

Composition and Injectivity

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. Suppose that $f \circ g$ is one-to-one. Prove that g must also be one-to-one.

$\mathcal{A}$. To prove that g is one-to-one, suppose $g(x) = g(y)$ for some $x, y \in A$. Applying f to both sides gives

$$f(g(x)) = f(g(y)). \implies (f \circ g)(x) = (f \circ g)(y).$$

Since $f \circ g$ is one-to-one, $x = y$. Hence,

$$\boxed{g \text{ is one-to-one.}}$$

Composition and Injectivity

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. Suppose that $f \circ g$ is one-to-one. Prove that g must also be one-to-one.

$\mathcal{A}$. To prove that g is one-to-one, suppose $g(x) = g(y)$ for some $x, y \in A$. Applying f to both sides gives

$$f(g(x)) = f(g(y)). \implies (f \circ g)(x) = (f \circ g)(y).$$

Since $f \circ g$ is one-to-one, $x = y$. Hence,

$$\boxed{g \text{ is one-to-one.}}$$

Composition: An Important Theorem

Theorem

Let $g : A \to B$ and $f : B \to C$, and suppose that $f \circ g$ is a bijection. Then, g is **onto** if and only if f is **one-to-one**.

Proof. ($\Rightarrow$) Suppose g is onto. Let $b, b' \in B$ and suppose $f(b) = f(b')$.

- Since g is onto, there exist $x, x' \in A$ such that $g(x) = b, g(x') = b'$. Thus,

$$(f \circ g)(x) = f(b) = f(b') = (f \circ g)(x').$$

- Since $f \circ g$ is one-to-one, $x = x'$, and hence $b = b'$. Therefore, f is one-to-one.

($\Leftarrow$) Suppose f is one-to-one. Let $b \in B$.

- Since $f(b) \in C$ and $f \circ g$ is onto, there exists $x \in A$ such that $(f \circ g)(x) = f(b)$.
- Hence, $f(g(x)) = f(b)$. Since f is one-to-one, $g(x) = b$. Thus every $b \in B$ has a preimage under g, so g is onto.

Composition: An Important Theorem

Theorem

Let $g : A \to B$ and $f : B \to C$, and suppose that $f \circ g$ is a bijection. Then, g is **onto** if and only if f is **one-to-one.**

Proof. **(⇒)** Suppose g is onto. Let $b, b' \in B$ and suppose $f(b) = f(b')$.

- Since g is onto, there exist $x, x' \in A$ such that $g(x) = b, g(x') = b'$. Thus,

$$(f \circ g)(x) = f(b) = f(b') = (f \circ g)(x').$$

- Since $f \circ g$ is one-to-one, $x = x'$, and hence $b = b'$. Therefore, f is one-to-one.

(⇐) Suppose f is one-to-one. Let $b \in B$.

- Since $f(b) \in C$ and $f \circ g$ is onto, there exists $x \in A$ such that $(f \circ g)(x) = f(b)$.
- Hence, $f(g(x)) = f(b)$. Since f is one-to-one, $g(x) = b$. Thus every $b \in B$ has a preimage under g, so g is onto.

Composition: An Important Theorem

Theorem

Let $g : A \to B$ and $f : B \to C$, and suppose that $f \circ g$ is a bijection. Then, g is **onto** if and only if f is **one-to-one.**

Proof. **($\Rightarrow$)** Suppose g is onto. Let $b, b' \in B$ and suppose $f(b) = f(b')$.

- Since g is onto, there exist $x, x' \in A$ such that $g(x) = b, g(x') = b'$. Thus,
$$(f \circ g)(x) = f(b) = f(b') = (f \circ g)(x').$$
- Since $f \circ g$ is one-to-one, $x = x'$, and hence $b = b'$. Therefore, f is one-to-one.

($\Leftarrow$) Suppose f is one-to-one. Let $b \in B$.

- Since $f(b) \in C$ and $f \circ g$ is onto, there exists $x \in A$ such that $(f \circ g)(x) = f(b)$.
- Hence, $f(g(x)) = f(b)$. Since f is one-to-one, $g(x) = b$. Thus every $b \in B$ has a preimage under g, so g is onto.

Composition: An Important Theorem

Theorem

Let $g : A \to B$ and $f : B \to C$, and suppose that $f \circ g$ is a bijection. Then, g is **onto** if and only if f is **one-to-one**.

Proof. (⇒) Suppose g is onto. Let $b, b' \in B$ and suppose $f(b) = f(b')$.

- Since g is onto, there exist $x, x' \in A$ such that $g(x) = b, g(x') = b'$. Thus,

$$(f \circ g)(x) = f(b) = f(b') = (f \circ g)(x').$$

- Since $f \circ g$ is one-to-one, $x = x'$, and hence $b = b'$. Therefore, f is one-to-one.

(⇐) Suppose f is one-to-one. Let $b \in B$.

- Since $f(b) \in C$ and $f \circ g$ is onto, there exists $x \in A$ such that $(f \circ g)(x) = f(b)$.
- Hence, $f(g(x)) = f(b)$. Since f is one-to-one, $g(x) = b$. Thus every $b \in B$ has a preimage under g, so g is onto.

Quick Check: A Counterexample

$\mathcal{Q}$. Find functions $g : A \to B$ and $f : B \to C$ such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.

$\mathcal{A}$. Consider

$$A = \{1, 2\}, \qquad B = \{a, b, c\}, \qquad C = \{x, y\}.$$

- Define $g : A \to B$ by $g(1) = a, \qquad g(2) = b.$
- Thus, g is **not onto**, since c is not in its range.
- Define $f : B \to C$ by $f(a) = x, \qquad f(b) = y, \qquad f(c) = x.$

Thus, f is **not one-to-one**, since $f(a) = f(c) = x$.

However, $(f \circ g)(1) = f(a) = x$, $(f \circ g)(2) = f(b) = y$. Therefore,

$$f \circ g : \begin{cases} 1 \mapsto x, \\ 2 \mapsto y, \end{cases}$$

which is both one-to-one and onto.

Quick Check: A Counterexample

$\mathcal{Q}$. Find functions $g : A \to B$ and $f : B \to C$ such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.

$\mathcal{A}$. Consider

$$A = \{1, 2\}, \qquad B = \{a, b, c\}, \qquad C = \{x, y\}.$$

- Define $g : A \to B$ by $g(1) = a, \qquad g(2) = b.$
- Thus, g is **not onto**, since c is not in its range.
- Define $f : B \to C$ by $f(a) = x, \qquad f(b) = y, \qquad f(c) = x.$

Thus, f is **not one-to-one**, since $f(a) = f(c) = x$.

However, $(f \circ g)(1) = f(a) = x, (f \circ g)(2) = f(b) = y$. Therefore,

$$f \circ g : \begin{cases} 1 \mapsto x, \\ 2 \mapsto y, \end{cases}$$

which is both one-to-one and onto.

Quick Check: A Counterexample

$\mathcal{Q}$. Find functions $g : A \to B$ and $f : B \to C$ such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.

$\mathcal{A}$. Consider

$$A = \{1, 2\}, \qquad B = \{a, b, c\}, \qquad C = \{x, y\}.$$

- Define $g : A \to B$ by $g(1) = a, \qquad g(2) = b.$
- Thus, g is **not onto**, since c is not in its range.
- Define $f : B \to C$ by $f(a) = x, \qquad f(b) = y, \qquad f(c) = x.$

Thus, f is **not one-to-one**, since $f(a) = f(c) = x$.

However, $(f \circ g)(1) = f(a) = x$, $(f \circ g)(2) = f(b) = y$. Therefore,

$$f \circ g : \begin{cases} 1 \mapsto x, \\ 2 \mapsto y, \end{cases}$$

which is both one-to-one and onto.

Quick Check: A Counterexample

$\mathcal{Q}$. Find functions $g : A \to B$ and $f : B \to C$ such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.

$\mathcal{A}$. Consider

$$A = \{1, 2\}, \qquad B = \{a, b, c\}, \qquad C = \{x, y\}.$$

- Define $g : A \to B$ by $g(1) = a, \qquad g(2) = b.$
- Thus, g is **not onto**, since c is not in its range.
- Define $f : B \to C$ by $f(a) = x, \qquad f(b) = y, \qquad f(c) = x.$

Thus, f is **not one-to-one**, since $f(a) = f(c) = x$.

However, $(f \circ g)(1) = f(a) = x, (f \circ g)(2) = f(b) = y$. Therefore,

$$f \circ g : \begin{cases} 1 \mapsto x, \\ 2 \mapsto y, \end{cases}$$

which is both one-to-one and onto.

Quick Check: A Counterexample

$\mathcal{Q}$. Find functions $g : A \to B$ and $f : B \to C$ such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.

$\mathcal{A}$. Consider

$$A = \{1, 2\}, \qquad B = \{a, b, c\}, \qquad C = \{x, y\}.$$

- Define $g : A \to B$ by $g(1) = a, \qquad g(2) = b.$
- Thus, g is **not onto**, since c is not in its range.
- Define $f : B \to C$ by $f(a) = x, \qquad f(b) = y, \qquad f(c) = x.$

Thus, f is **not one-to-one**, since $f(a) = f(c) = x$.

However, $(f \circ g)(1) = f(a) = x, (f \circ g)(2) = f(b) = y$. Therefore,

$$f \circ g : \begin{cases} 1 \mapsto x, \\ 2 \mapsto y, \end{cases}$$

which is both one-to-one and onto.

Quick Check: A Counterexample

$\mathcal{Q}$. Find functions $g : A \to B$ and $f : B \to C$ such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.

$\mathcal{A}$. Consider

$$A = \{1, 2\}, \qquad B = \{a, b, c\}, \qquad C = \{x, y\}.$$

- Define $g : A \to B$ by $g(1) = a, \qquad g(2) = b.$
- Thus, g is **not onto**, since c is not in its range.
- Define $f : B \to C$ by $f(a) = x, \qquad f(b) = y, \qquad f(c) = x.$

Thus, f is **not one-to-one**, since $f(a) = f(c) = x$.

However, $(f \circ g)(1) = f(a) = x, (f \circ g)(2) = f(b) = y$. Therefore,

$$f \circ g : \begin{cases} 1 \mapsto x, \\ 2 \mapsto y, \end{cases}$$

which is both one-to-one and onto.

Quick Check: A Counterexample

$\mathcal{Q}$. Find functions $g : A \to B$ and $f : B \to C$ such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.

$\mathcal{A}$. Consider

$$A = \{1, 2\}, \qquad B = \{a, b, c\}, \qquad C = \{x, y\}.$$

- Define $g : A \to B$ by $g(1) = a, \qquad g(2) = b.$
- Thus, g is **not onto**, since c is not in its range.
- Define $f : B \to C$ by $f(a) = x, \qquad f(b) = y, \qquad f(c) = x.$

Thus, f is **not one-to-one**, since $f(a) = f(c) = x$.

However, $(f \circ g)(1) = f(a) = x$, $(f \circ g)(2) = f(b) = y$. Therefore,

$$f \circ g : \begin{cases} 1 \mapsto x, \\ 2 \mapsto y, \end{cases}$$

which is both one-to-one and onto.

Quick Check: Injectivity of a Composition

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? Justify your answer.

$\mathcal{A}$. **Yes.** In fact, we do not even need to assume that f is one-to-one.

- Suppose $g(x) = g(y)$ for some $x, y \in A$.
- Applying f to both sides gives $f(g(x)) = f(g(y))$. Therefore,

$$(f \circ g)(x) = (f \circ g)(y).$$

- Since $f \circ g$ is one-to-one, $x = y$. Hence,

$$\boxed{g \text{ is one-to-one.}}$$

Quick Check: Injectivity of a Composition

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? Justify your answer.

$\mathcal{A}$. **Yes.** In fact, we do not even need to assume that f is one-to-one.

- Suppose $g(x) = g(y)$ for some $x, y \in A$.
- Applying f to both sides gives $f(g(x)) = f(g(y))$. Therefore,

$$(f \circ g)(x) = (f \circ g)(y).$$

- Since $f \circ g$ is one-to-one, $x = y$. Hence,

$$\boxed{g \text{ is one-to-one.}}$$

Quick Check: Surjectivity of a Composition

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. If f and $f \circ g$ are onto, does it follow that g is onto? Justify your answer.

$\mathcal{A}$. **No.** Consider

$$A = \{1\}, \qquad B = \{a, b\}, \qquad C = \{x\}.$$

- Define $g(1) = a$ and $f(a) = x, \qquad f(b) = x$. Then f is onto because x has a preimage.
- Also, $(f \circ g)(1) = f(a) = x$, so $f \circ g$ is onto.

However, g is **not onto**, since $b \in B$ has no preimage under g.

Thus,

$$\boxed{f \text{ onto and } f \circ g \text{ onto} \;\not\Rightarrow\; g \text{ onto.}}$$

Quick Check: Surjectivity of a Composition

Let $g : A \to B$ and $f : B \to C$.

$\mathcal{Q}$. If f and $f \circ g$ are onto, does it follow that g is onto? Justify your answer.

$\mathcal{A}$. **No.** Consider

$$A = \{1\}, \qquad B = \{a, b\}, \qquad C = \{x\}.$$

- Define $g(1) = a$ and $f(a) = x, \qquad f(b) = x$. Then f is onto because x has a preimage.
- Also, $(f \circ g)(1) = f(a) = x$, so $f \circ g$ is onto.

However, g is **not onto**, since $b \in B$ has no preimage under g.

Thus,

$$\boxed{f \text{ onto and } f \circ g \text{ onto } \not\Rightarrow \; g \text{ onto.}}$$

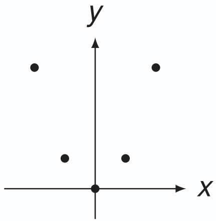
Graphs of Functions

Definition

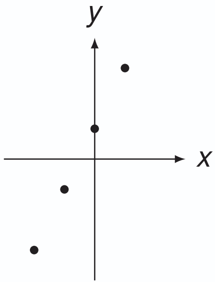
Let $f : A \to B$ be a function. The **graph** of f is the set of ordered pairs:

$$\{(a, b) \mid a \in A \wedge f(a) = b\}$$

Graph of $f(x) = x^2$ from $\mathbb{Z}$ to $\mathbb{Z}$:



Graph of $f(n) = 2n + 1$ from $\mathbb{Z}$ to $\mathbb{Z}$:



Thank You for your attention!

Any questions?